



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Artificial Intelligence and Data Science

Academic Year: 2024- 2025 (Odd Semester)

Active Learning Best practices: Flash Card - Cooperative Learning Technique

Degree/Branch : B.Tech. / Artificial Intelligence and Data Science
Year/Semester & Section : II Year / III Semester & “A” Section
Course Code & Title : AD3351 – Design and Analysis of Algorithms
Name of the Faculty-Member : Mrs.B.SankaraLakshmi, AP/AI&DS

Theme of Discussion : Huffman Trees and Codes
Topics Covered : Unit - III
Date and Time : 28.10.2024 & 03.45 PM to 04.30 PM

Course Outcome : CO3
Topic Learning Outcome : 27
Activity Chosen : Flash Card

Active Learning Best practices: Flash Card - Cooperative Learning Technique

Topic : Huffman Trees and Codes



Greedy Technique

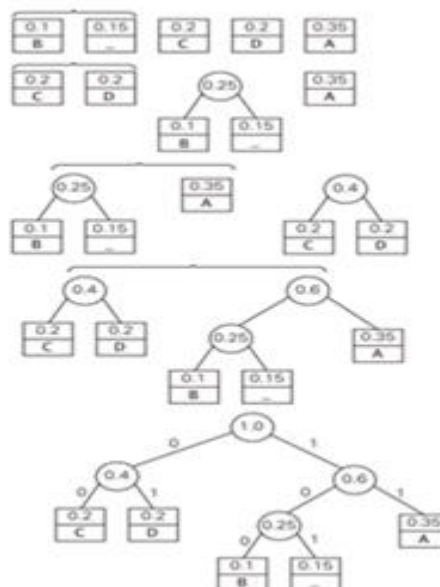


FIGURE 9.12: Example of constructing a Huffman coding tree.

Learning Outcomes

a. Active Recall

Flashcards help students engage in active recall, which is the process of stimulating memories and creating lasting connections to material.

b. Memory Retention

Flashcards can help with memory retention through repetition. Repetition is especially important in the early stages of life, when the brain is better able to process and store information.

c. Spaced repetition

Flashcards can help with spaced repetition, which can help you learn twice as efficiently as other study methods.

Procedure



Step 1 (15 minutes)

- Students spend more time on the preparation of the topic given earlier.
- Go through the topic in a deep manner and understand the concepts well.

Step 2 (10 minutes)

- Students have a discussion among themselves on the specific topic given to improve their communication skill.
- Share the knowledge gained with each other to analyze their listening ability.
- Identify some important points and the differences relevant to the topic.

Step 3 (10 minutes)

- Students highlight some points in the text whichever they have identified through the discussion.
- Explain each highlighted point to others to implement the knowledge sharing.

Step 4 (10 minutes)

- Students make flash cards on the topic.
- Flash Cards must not be too lengthy or too short.
- Flash Cards are made with clarified concepts of the topic along with an example to improve the problem-solving skill.

Design and Implementation

- Procedure of Flash Card has been discussed in the course website in advance in which students knew the knowledge of this activity.

Glimpses:

Huffman tree and codes:

* The Huffman code is a lossless data compression technique codes to the data represent the tree and codes to the characters are frequently in the encoding.

Comparison of Huffman and ASCII

ASCII code: To get a fixed characters to compare the Huffman code to a code bit is represent.

Huffman code: To represent the characters it have a data compression codes.

Benefits:

- Efficient:** To represent the data to the efficient of the each bits.
- Lossless:** No characters are lost the encoding and the decoding.

Eg:

D A D
01 11 10

Huffman codes and trees:

This is a lossless or non-loss method of converting the symbols or variables into bits which are to be encoded and decoded.

- Here each word which has higher frequency gets the minimum Binary value when compared to the minimum frequency symbols.
- A Binary tree is generated according to the frequency values like A=0.35, B=0.1 (Here A is most occurring so gets the minimum value when compared to B).

Comparison with ASCII codes:

- ASCII codes has fixed length of 7-8 bits for all the symbols and characters.
- Huffman codes has variable length of bits according to the frequency it occurs or no. of times it occurs.

Benefits:

- Efficient compression:** Here the bit values are assigned according to the no. of times the value occurs. Hence the codes are compressed when compared to ASCII codes.
- Non-loss or lossless:** No data is lost or missed and all the elements are encoded and decoded.

Example:

This is the format of binary tree generated by Huffman codes.

Virudhunagar, Tamil Nadu, India
 Ramco Institute of Technology Road, Rajapalayam,
 Virudhunagar, 626117, Tamil Nadu, India
 Lat 9.482931, Long 77.514053
 10/28/2024 04:39 PM GMT+05:30

Huffman Trees and Codes

Huffman tree works by building the binary tree called Huffman tree. It construct is based on frequency of occurrence of each character in binary tree. If the tree has highest frequency one closer to the root of the tree and have lowest frequency further away from the root of tree.

Benefits:

- Huffman tree is very efficient this compress are quick and efficient.
- It is relatively simple to implement.
- Huffman tree can use any datatype.
- It is efficient for the large files.

Comparison of ASCII and Huffman tree

ASCII CODE	HUFFMAN CODE
1. It is a fixed-length, it never changes for each character the be same.	1. It is a Variable-length, it vary the length for each character.

HUFFMAN TREES AND CODES

Huffman Tree and codes is a data compression technique and algorithm.

- Encoding - Sequence of bits**
- 2 types of Huffman Tree and codes, they are:
 - Fixed length Encoding - assign strings of bits in a same length.
 - Variable length Encoding - assign strings of bits in a different symbol and different length.

Problem: 1st symbol does not know how many bits required to encode things we use prefix-free codes or free-free codes.

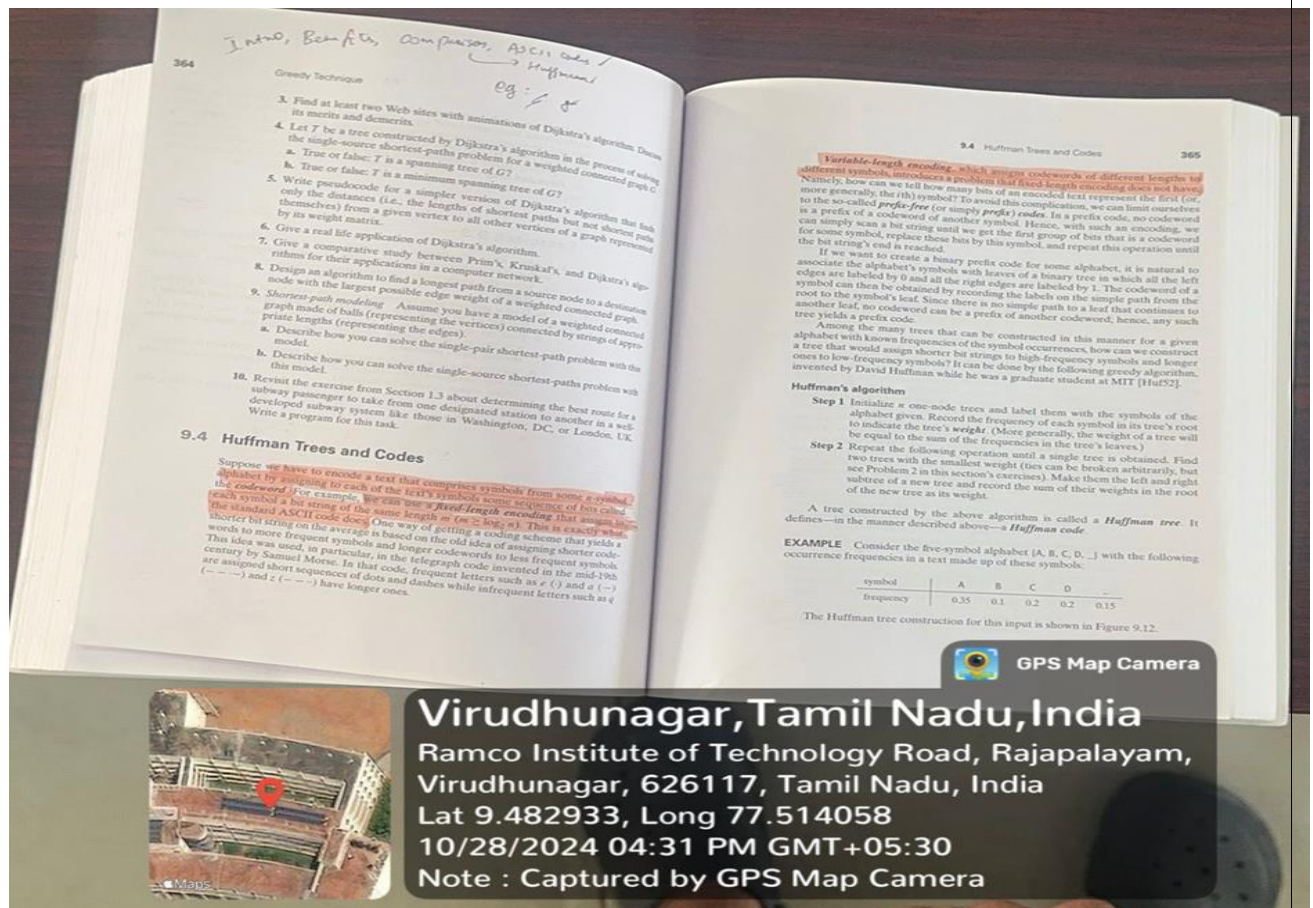
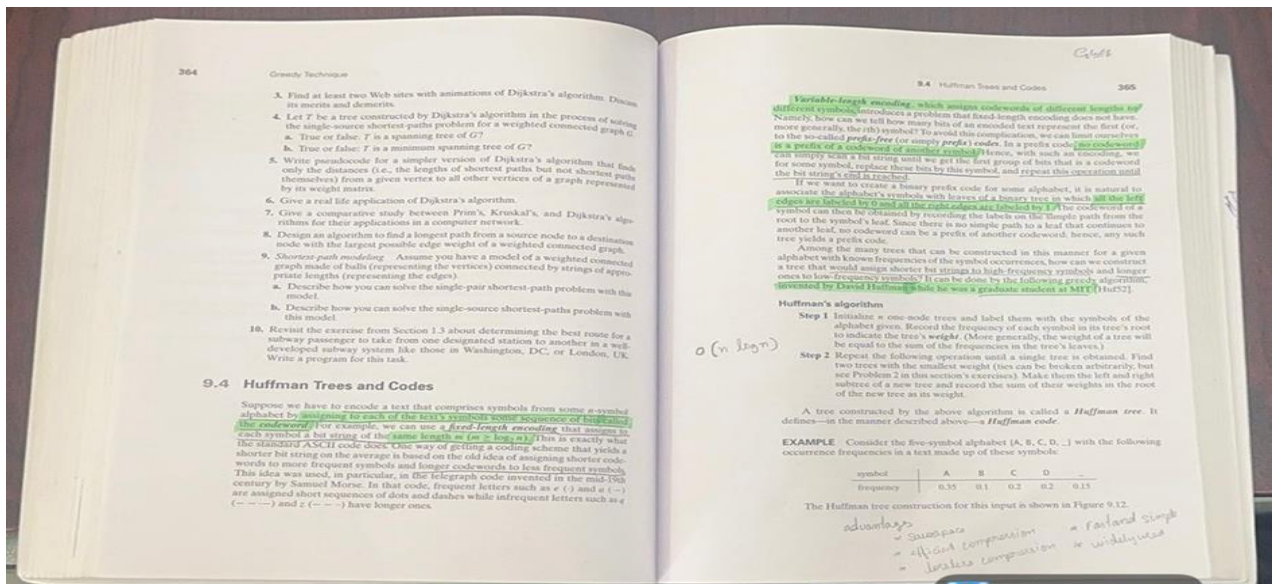
Prefix code word: No code word is prefix code word of another symbol.

- Left side of the tree labelled as '0' and right side of the tree labelled as '1'.

Benefits:

- It effective and efficient way for large data files.
- Non-loss compression.
- Adaptability.
- Fast decoding.

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 Note: Captured by GPS Map Camera





Reflective Report

Challenges and Strategies

- Make all the students to bring to attention towards participation in the activity.
- Motivate the slow learners to involve the activity.
- Specific topic was already given in the previous classes to lead the activity effectively.
- Make the students to memorize the facts, definitions and or short pieces of information.

Observations

Flash Cards are a versatile and an effective study tool. Using flashcards is one of the most common study method, and it is the one many students are most familiar with. They are an active learning technique that helps improve recall ability. Flashcards help students to engage in active recall or a process wherein students actively engage in learning by stimulating our memories and creating lasting connections to the material. Flash cards are a very useful revision activity for many reasons. They work across all subjects; they can be used with the recall of facts, topics, keywords and more. They are a very simple technique for learners to use – low effort but high impact.

Students Response

- Bright students were actively participated to prepare the flash cards based on the concepts.
- Slow learners have understood the highlighted points well by having discussion.
- Faculty has involved in making flash cards.
- Most of the students were effectively utilized the session conducted to learn more.

Students response ensures the students could improve listening, communication, creativity and problem-solving skills.



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Feedback collected in the class and also through online

Google form link : <https://forms.gle/9423mGSLH56PTuP77>

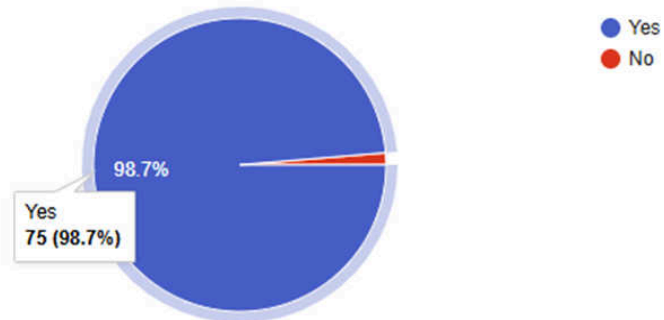
Feedback Questionnaire:

- Did the plan align with your expectations?
Yes No
- How effectively was the Innovative Practice Plan implemented?
Excellent Good Satisfactory
- How well did you understand the Objectives and Purpose of the Innovative Practice Plan?
1 2 3 4 5
- Were the necessary resources and support provided for successful implementation?
Yes No
- Did you notice any specific benefits or changes resulting from this plan?
Yes No
- How were you engaged throughout the practice plan?
1 2 3 4 5
- Did the plan provide opportunities for Active Participation and Collaboration?
Yes No
- Whether the topic Huffman Trees and Codes has been analyzed in a best way using this Innovative Practice Plan?
Excellent Good Satisfactory
- Does this active learning improve listening, thinking describing and problem-solving skills?
Yes No
- Is the practice plan sufficient to understand and discuss the concepts?
Yes No

Feedback Analysis:

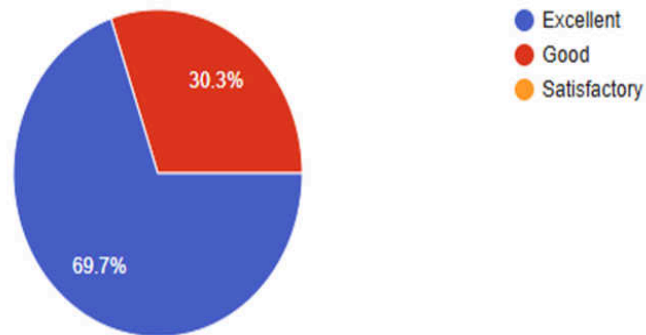
1. Did the plan align with your expectations?

76 responses



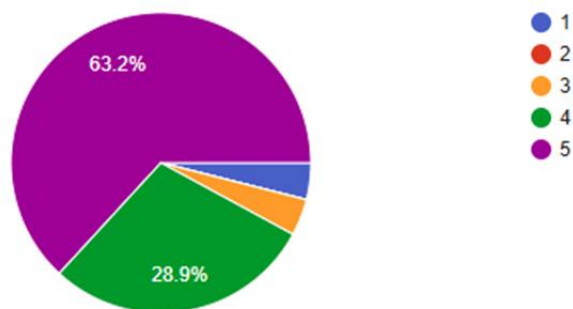
2. How effectively was the Innovative Practice Plan implemented?

76 responses



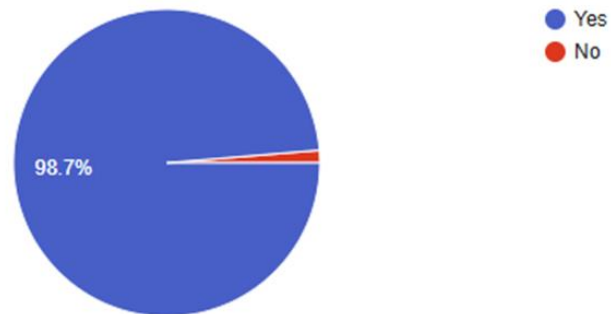
3. How well did you understand the Objectives and Purpose of the Innovative Practice Plan?

76 responses



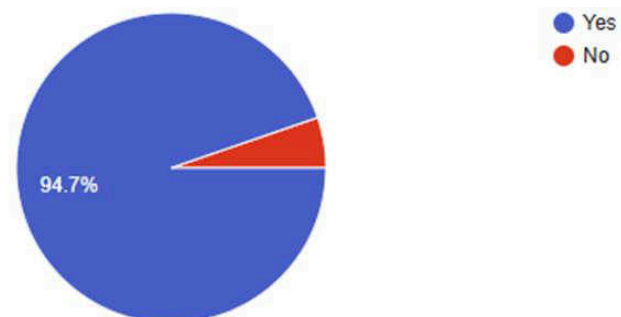
4. Were the necessary resources and support provided for successful implementation?

76 responses



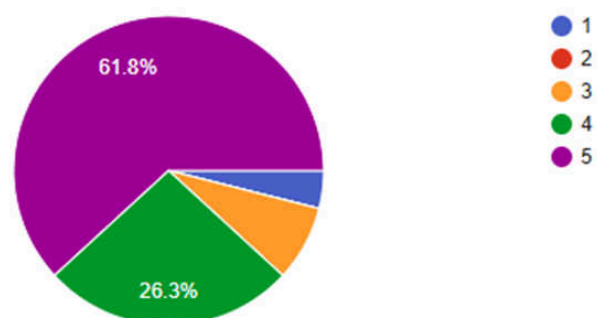
5. Did you notice any specific Benefits or Changes resulting from this plan?

76 responses



6. How were you engaged throughout the Practice Plan?

76 responses



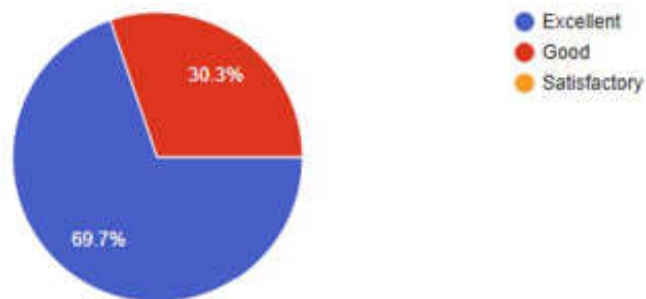
7. Did the plan provide Opportunities for Active Participation and Collaboration?

76 responses



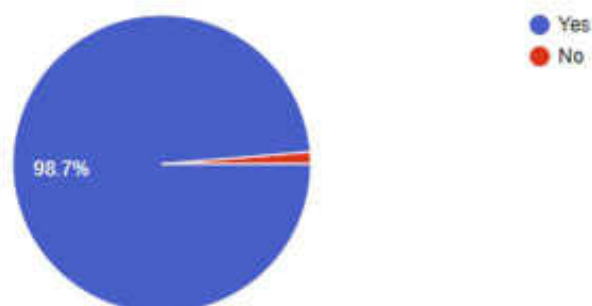
8. Whether the topic Huffman Trees and Codes has been analysed in a best way using this Innovative Practice Plan?

76 responses



9. Does this activity learning improve Listening, Thinking, Describing and Problem-Solving skills?

76 responses



10. Is the Practice Plan sufficient to understand and discuss the concepts?

76 responses



A handwritten signature in black ink, appearing to be 'S. S. S.', with the date '30/10/24' written below it.

Signature of Faculty Member

A handwritten signature in black ink, appearing to be 'mah', with the date '30/10' written below it.

HOD